

Nonlinear simulation of frequency chirping of hot electron interchange mode in a dipole-confined plasma

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A source-driven simulation of the hot electron interchange (HEI) instability in a dipole-confined plasma is presented using a vorticity approach. Unlike previous fixed-profile simulations, this work introduces a sustained hot electron source, allowing the profiles to evolve self-consistently towards a nonlinear steady state. We derive the vorticity equation governing the HEI instability and develop a code that enables nonlinear simulation of multi-energy-group hot electron dynamics. The simulation results demonstrate that under sustained source driving, the system spontaneously forms phase-space hole structures accompanied by upward frequency chirping. The vorticity approach developed in this work provides a more efficient tool for understanding energetic particle transport and nonlinear wave-particle interactions in dipole geometry.

I. INTRODUCTION

The closed-field-line dipole configuration is relevant to the study of planetary magnetospheres [1–4], magnetic fusion confinement [5–8], and magnetically confined non-neutral plasmas [9–11]. The hot electron interchange instabilities (HEI), first described by Krall [12], play a key role in the radial transport of energetic electrons in the magnetic mirror and dipole configurations. These energetic particle instabilities represent interesting and important physics. The HEI has been extensively studied in the laboratory experiments, particularly in the CTX [13] and the LDX [14–16]. In experiments, the hot electrons are equatorially trapped near the microwave cyclotron resonance, producing a steep radial gradient in the hot electron number density profiles. Consequently, a population of hot electrons trapped in the dipole field can excite low-frequency, flute-type electrostatic fluctuations through drift-resonant wave-particle interactions, leading to cross-field transport and the formation of complex phase-space structures [17, 18].

Observations in CTX demonstrate that the HEI instability driven by hot electrons exhibits rich nonlinear behavior, including frequency sweeping and the spontaneous formation of phase-space holes [1, 19]. Frequency sweeping can be suppressed by applying radio frequency fields, which is attributed to the destruction of phase-space holes by localized velocity-space diffusion [20, 21]. Furthermore, a fully self-consistent model for the nonlinear evolution of the HEI has been developed [1, 19]. In this model, the potential fluctuations are computed by Poisson equation, which gives an elliptic equation for the time derivative of the potential in terms of the $\mathbf{E} \times \mathbf{B}$ convection of the fluctuating charge density, the divergence of the nonaxisymmetric ion polarization currents, and the magnetic drift of the hot electrons [1, 19]. However, this approach requires evolving the ion conti-

nunity equation and using iterative methods for the elliptic equation, resulting in significant computational cost [19]. The previous simulations of HEI based on solving Poisson equation have demonstrated the inward motion of phase-space holes [17] and the rising frequency of the instability [1, 19].

For the experiments in CTX, both the observations and the numerical simulations appear to be explained by a generalized nonlinear theory for weakly unstable plasmas developed by Berk, Breizman, and co-workers [22, 23]. In this theory, at saturation, the wave frequencies continue to change as energy nonlinearly extracted from the resonant particles is balanced by nonresonant dissipation, suggesting frequency sweeping to be a general nonlinear wave-particle effect.

In this work, we present a self-consistent, nonlinear model based on the vorticity equation for the evolution of a quasineutral plasma consisting of cold ions and a mixture of hot and cold electrons confined by a dipole magnetic field. The vorticity model explores the flute-like electrostatic response to a steep radial gradient of hot electrons confined in a dipole geometry. The nonlinear simulations describe the self-consistent dynamics of the hot electrons and the growth and saturation of the low-frequency HEI. This description is relatively simple because the low-frequency electrostatic instabilities are flute-like, and the dynamics of the deeply trapped hot electrons are two-dimensional. A source term for hot electron production adds to the distribution evolution equation, which allows the hot electron profile to evolve self-consistently. Previously, a conserving source term has been included in the ion continuity equation for simulating quasi-steady driven interchange turbulence [24].

This paper is organized as follows. Section II presents a brief description of the basic equations in dipole geometry. The vorticity equation for the quasineutrality is introduced in Sec. III. The linear stability of the HEI is described in Sec. IV. In Sec. V the fully nonlinear simu-

lations of the HEI are presented. In the final section, we summarize key conclusions.

II. BASIC EQUATIONS

In the following, we summarize a self-consistent, non-linear model for the flute-like electrostatic, low-frequency HEI in dipole magnetic flux coordinates. The axisymmetric dipole field $\mathbf{B} = \nabla\varphi \times \nabla\psi = \nabla\chi$ where χ is the coordinate along the field line, ψ is the magnetic flux and φ is the azimuthal angle. The model describes a plasma comprising cold ions and a mix of cold and hot electrons. Particle motion is confined to trajectories in a two-dimensional phase-space, (ψ, φ) [1, 25].

For low-frequency fluctuations ($\omega \ll \omega_b$), the hot electron population is described by a bounce-averaged distribution function $F = \langle f \rangle_b$ [25]. The number of particles per unit magnetic flux is defined by

$$N(\psi, \varphi) = \int \frac{d\chi}{B^2} \iint d^3v F. \quad (1)$$

The velocity integral can be written as

$$\iint d^3v = \frac{2\pi B}{m^2} \sum_{+-} \iint \frac{d\mu dE}{|v_{\parallel}|}, \quad (2)$$

where the kinetic energy $E = m(v_{\perp}^2 + v_{\parallel}^2)/2$ and magnetic moment $\mu = mv_{\perp}^2/(2B)$, and $\sum_{+-}$ represents the sum over the positive and negative parallel velocity. Since F is the bounce-averaged distribution function, Eq. (1) can be written

$$N = \frac{2\pi}{m^2} \iint d\mu dE \left(\sum_{+-} \int \frac{d\chi}{B} \frac{1}{|v_{\parallel}|} \right) F, \quad (3)$$

where the sum over the field-line integral defines the full bounce period:

$$\tau_b = \oint \frac{dl}{|v_{\parallel}|} = \sum_{+-} \int \frac{d\chi}{B} \frac{1}{|v_{\parallel}|}. \quad (4)$$

Thus we find

$$N = \frac{2\pi}{m^2} \iint \tau_b dE d\mu F. \quad (5)$$

Consider the longitudinal invariant

$$J = \oint dl |v_{\parallel}| = \oint dl \sqrt{\frac{2}{m}(E - \mu B(l))}, \quad (6)$$

where $B(l)$ is the magnetic field along the field line. Changing the integration variable from E to J at con-

stant μ , we use

$$dJ = \left(\frac{\partial J}{\partial E} \right)_{\mu} dE = \frac{\tau_b}{m} dE. \quad (7)$$

Then Eq. (5) becomes

$$N = \frac{2\pi}{m} \iint dJ d\mu F. \quad (8)$$

The evolution of hot electron distribution function is [25]

$$\frac{\partial F}{\partial t} + \frac{\partial}{\partial \varphi} [(\omega_d + v_{\phi}) F] + \frac{\partial}{\partial \psi} (v_{\psi} F) = 0, \quad (9)$$

where the plasma motions due to electrostatic potential fluctuations are

$$v_{\phi} = -\frac{\partial \Phi}{\partial \psi}, \quad v_{\psi} = \frac{\partial \Phi}{\partial \varphi}. \quad (10)$$

For the dipole field strength $B \propto \psi^3$, the magnetic drift frequency of hot electrons is given by

$$\omega_d = \frac{\mu}{e} \frac{\partial B}{\partial \psi} = \frac{3\mu B}{e\psi}. \quad (11)$$

The field-line averaged Poisson equation is

$$\int \frac{d\chi}{B^2} \nabla_{\perp}^2 \Phi = -4\pi e \int \frac{d\chi}{B^2} (n_i - n_e). \quad (12)$$

where the electrostatic potential Φ is assumed to be constant along the dipole field line. In dipole magnetic flux coordinates, the flux-tube integral of the divergence of a quantity $\mathbf{A}$ is given by [25]

$$\int \nabla \cdot \mathbf{A} \frac{d\chi}{B^2} = \frac{\partial}{\partial \psi} \int \nabla \psi \cdot \mathbf{A} \frac{d\chi}{B^2} + \frac{\partial}{\partial \varphi} \int \nabla \varphi \cdot \mathbf{A} \frac{d\chi}{B^2}. \quad (13)$$

Axial symmetry implies $|\nabla \varphi| = 1/R$ and $|\nabla \psi| = RB$ with R the major radius from the dipole axis. Using Eq. (13) and $\nabla_{\perp} \Phi = (\partial \Phi / \partial \psi) \nabla \psi + (\partial \Phi / \partial \varphi) \nabla \varphi$, we find

$$\int \frac{d\chi}{B^2} \nabla^2 \Phi = \frac{\partial}{\partial \psi} \int \frac{d\chi}{B^2} R^2 B^2 \frac{\partial \Phi}{\partial \psi} + \frac{\partial}{\partial \varphi} \int \frac{d\chi}{B^2} \frac{1}{R^2} \frac{\partial \Phi}{\partial \varphi}. \quad (14)$$

For a point dipole, the geometric integrals along the field line can be obtained (see Appendix). Then Eq. (12) becomes [1]

$$\frac{\partial}{\partial \varphi} \left(2 \frac{M}{\psi^2} \frac{\partial \Phi}{\partial \varphi} \right) + \frac{\partial}{\partial \psi} \left(4M \frac{\partial \Phi}{\partial \psi} \right) = -4\pi e (N_i - N_e), \quad (15)$$

where N_i and N_e are defined as the particle numbers per unit flux.

The evolution of cold ion density is described by the continuity equation $\partial n / \partial t + \nabla \cdot (n \mathbf{v}) = 0$, where $\mathbf{v}$ is the sum of $\mathbf{E} \times \mathbf{B}$ and polarization drifts. Use of Eq. (13)

yields the field-line averaged $\mathbf{E} \times \mathbf{B}$ convection term

$$\int \frac{d\chi}{B^2} \nabla \cdot (n\mathbf{v}_E) = \frac{\partial}{\partial\psi}(Nv_\psi) + \frac{\partial}{\partial\varphi}(Nv_\varphi) \quad (16)$$

With Eq. (13) we obtain the flux-tube integrated polarization drift term

$$\begin{aligned} & \int \frac{d\chi}{B^2} \nabla \cdot \left(\frac{nM_i}{B^2} \frac{\partial}{\partial t} \nabla \Phi \right) \\ &= \frac{\partial}{\partial\varphi} \left(N_i \Upsilon_\varphi \frac{\partial}{\partial\varphi} \dot{\Phi} \right) + \frac{\partial}{\partial\psi} \left(N_i \Upsilon_\psi \frac{\partial}{\partial\psi} \dot{\Phi} \right) \end{aligned} \quad (17)$$

where the dot denotes the time derivative $\dot{\Phi} = \partial\Phi/\partial t$ and the density-weighted field line averages [25, 26]

$$\Upsilon_\varphi = \frac{1}{N} \int \frac{d\chi}{B^2} \frac{nM_i}{B^2} \frac{1}{R^2}, \quad \Upsilon_\psi = \frac{1}{N} \int \frac{d\chi}{B^2} \frac{nM_i}{B^2} R^2 B^2 \quad (18)$$

Thus the continuity equation for cold ions is [1]

$$\begin{aligned} & \frac{\partial N_i}{\partial t} + \frac{\partial}{\partial\psi}(N_i v_\psi) + \frac{\partial}{\partial\varphi}(N_i v_\varphi) \\ &= \frac{\partial}{\partial\varphi}(N_i u_\varphi) + \frac{\partial}{\partial\psi}(N_i u_\psi) \end{aligned} \quad (19)$$

where

$$u_\varphi = \Upsilon_\varphi \frac{\partial^2 \Phi}{\partial\varphi \partial t}, \quad u_\psi = \Upsilon_\psi \frac{\partial^2 \Phi}{\partial\psi \partial t} \quad (20)$$

is the linear part of polarization drift. The nonlinear advective part in the polarization drift has been neglected.

III. DERIVATION OF VORTICITY EQUATION

We now derive the dimensionless form of the basic equations. The magnetic field and density are normalized to their values at an equatorial reference radius L_0 . The magnetic moment of hot electrons μ is normalized by a reference value μ_0 . The electrostatic potential Φ is normalized to $3\mu_0 B_0/e$. The time is normalized to $1/\omega_{d0}$ where

$$\omega_{d0} = \frac{3\mu_0 B_0}{e\psi_0} \quad (21)$$

is the characteristic magnetic drift frequency for hot electrons. We denote $x = \psi/\psi_0$ as the normalized magnetic flux and $y = \varphi$ as the azimuthal angle, where ψ_0 is the magnetic flux at the equatorial reference radius.

Then the normalized Eq. (9) for the evolution of hot electron distribution function is

$$\frac{\partial F}{\partial t} + \frac{\partial}{\partial y} [(\hat{\omega}_d + v_y)F] + \frac{\partial}{\partial x}(v_x F) = 0, \quad (22)$$

where

$$\hat{\omega}_d = \frac{\omega_d}{\omega_{d0}} = \mu x^2, \quad (23)$$

$$v_y = -\frac{\partial\phi}{\partial x}, \quad v_x = \frac{\partial\phi}{\partial y} \quad (24)$$

For deeply trapped hot electrons with $J = 0$, the distribution reduces to $F = F(\mu, \psi, \varphi, t) \delta(J)$. Integrating over J and absorbing the constant $2\pi/m$ into F , we obtain the hot electron number density

$$N_h(x, y, t) = \int d\mu F(x, y, t, \mu). \quad (25)$$

Integrating Eq. (22) over μ yields the hot electron continuity equation:

$$\frac{\partial}{\partial t} N_h + \frac{\partial}{\partial y} \left(v_y N_h + \int d\mu \hat{\omega}_d F \right) + \frac{\partial}{\partial x} (v_x N_h) = 0. \quad (26)$$

Cold electrons with $\mu = 0$ execute radial and azimuthal $\mathbf{E} \times \mathbf{B}$ motion

$$\frac{\partial}{\partial t} N_c + \frac{\partial}{\partial y} (v_y N_c) + \frac{\partial}{\partial x} (v_x N_c) = 0. \quad (27)$$

The total electron density is $N_e = N_c + N_h$. Adding Eqs. (26) and (27) gives the electron continuity equation

$$\dot{N}_e + \frac{\partial}{\partial y} (v_y N_e) + \frac{\partial}{\partial x} (v_x N_e) + \frac{\partial}{\partial y} \int d\mu \hat{\omega}_d(\mu) F = 0. \quad (28)$$

The normalized cold ion evolution equation (19) is

$$\dot{N}_i + \frac{\partial}{\partial y} (v_y N_i) + \frac{\partial}{\partial x} (v_x N_i) = \frac{\omega_{d0}}{\omega_{ci0}} \left(\frac{\partial}{\partial y} N_i u_y + \frac{\partial}{\partial x} N_i u_x \right) \quad (29)$$

with the linear part of polarization drift for a point dipole geometry [25]

$$u_y = \frac{0.6}{x^4} \frac{\partial}{\partial y} \dot{\phi}, \quad u_x = \frac{0.7}{x^2} \frac{\partial}{\partial x} \dot{\phi} \quad (30)$$

For simplicity, the polarization drift term is linearized with $N_{i0} \simeq 1$ for the background density:

$$\frac{\partial}{\partial y} (N_i u_y) + \frac{\partial}{\partial x} (N_i u_x) \simeq \frac{0.6}{x^4} \frac{\partial^2}{\partial y^2} \dot{\phi} + \frac{0.7}{x^2} \frac{\partial^2}{\partial x^2} \dot{\phi} \quad (31)$$

The normalized Poisson equation (15) takes the form

$$\frac{2}{x^2} \frac{\partial^2}{\partial y^2} \phi + 4 \frac{\partial^2}{\partial x^2} \phi = 0.91 \left(\frac{L_0}{\lambda_D} \right)^2 (N_e - N_i) \quad (32)$$

Given the experimental parameters $(L_0/\lambda_D)^2 \sim 10^3 \gg 1$, the righthand side dominates, enforcing the quasineutrality condition $N_e = N_i$, or equivalently

$$\dot{N}_e - \dot{N}_i = 0 \quad (33)$$

With Eqs. (28) and (29) the quasineutrality condition (33) yields the vorticity equation:

$$\frac{\partial}{\partial y} \int d\mu \hat{\omega}_d F + \frac{1}{\hat{\omega}_{ci0}} \left(\frac{0.6}{x^4} \frac{\partial^2}{\partial y^2} + \frac{0.7}{x^2} \frac{\partial^2}{\partial x^2} \right) \dot{\phi} = 0, \quad (34)$$

where $\hat{\omega}_{ci0} = \omega_{ci0}/\omega_{d0}$. This vorticity equation couples the hot electron magnetic drift with the cold ion polarization drift, providing a description of the HEI dynamics.

IV. LINEAR HEI MODES

Because of axial symmetry in a dipole, the perturbations are expressed in Fourier series $\Phi(x, y, t) = \sum_m \Phi_m(x) \exp(imy - i\omega t)$, where m is the mode number in the azimuthal direction. From the linearized kinetic equation (22), we have the perturbed hot electron distribution function

$$\tilde{F} = \frac{m\tilde{\phi}}{\omega - m\hat{\omega}_d} \frac{\partial F_0}{\partial x} \quad (35)$$

where ω is the complex frequency, and F_0 is the equilibrium distribution function. When the electrostatic potential consists of azimuthally propagating waves, the wave-particle resonance condition is $\omega = m\hat{\omega}_d = m\mu x^2$, which relates a particle's energy μ to a resonant flux surface $x = \sqrt{\omega/m\mu}$.

The hot electron equilibrium distribution takes the form [1]

$$F_0(x, \mu) = \alpha N_{i0} G(\mu) \delta(J) \quad (36)$$

where $\alpha(x) = N_h/N_{i0}$ is the fractional density of hot electrons, $G(\mu)$ is the normalized magnetic moment distribution satisfying $\int G(\mu) d\mu = 1$, and $\delta(J)$ indicates that electrons are deeply trapped in the equatorial plane ($J = 0$). Taking the Fourier transform of Eq. (34) at $x \simeq 1$, we obtain

$$m \int d\mu \hat{\omega}_d \tilde{F} + \frac{1}{\hat{\omega}_{ci0}} m_{\perp}^2 \omega \phi = 0 \quad (37)$$

where $m_{\perp}^2 = 0.6m^2 + 0.7k_x^2$ and the wavenumber k_x represents the radial mode width. Substitution of Eqs. (35) and (36) into Eq. (37) yields

$$m^2 \frac{\partial(N_{i0}\alpha)}{\partial x} \int d\mu \frac{\hat{\omega}_d G(\mu)}{\omega - m\hat{\omega}_d} \phi + \frac{1}{\hat{\omega}_{ci0}} m_{\perp}^2 \omega \phi = 0 \quad (38)$$

When $G(\mu') = \delta(\mu' - \mu)$, the hot electrons are monoenergetic. Then Eq. (38) gives the simplified dispersion relation [1]:

$$\omega^2 - m\hat{\omega}_d \omega + \Gamma_h^2 = 0 \quad (39)$$

where $\hat{\omega}_d = \mu x^2$ is the hot electron drift frequency and $\Gamma_h^2 \equiv (m^2/m_{\perp}^2) \hat{\omega}_{ci0} \hat{\omega}_d \partial(N_{i0}\alpha)/\partial x$ is the hot electron in-

terchange drive.

Solving the quadratic equation (39) for the complex frequency ω we have

$$\omega = \frac{m\hat{\omega}_d}{2} + \frac{1}{2} \sqrt{m^2 \hat{\omega}_d^2 - 4\Gamma_h^2} \quad (40)$$

Thus the low-frequency HEI instability occurs when the hot electron density gradients exceed a stability threshold:

$$\frac{\partial(N_{i0}\alpha)}{\partial x} > \frac{\omega_d}{\omega_{ci0}} \frac{m_{\perp}^2}{4} \quad (41)$$

The HEI modes are more unstable with small perpendicular mode number $m_{\perp}$ (low values of m and broad radial mode structures of the electrostatic potential, $k_x \simeq 0$). The drive for the instability is the radial phase-space density, not the energy distribution. Thus the energetic electrons with a Maxwellian distribution and overly peaked radial profiles can also drive the HEI.

V. NONLINEAR SIMULATION OF THE HEI

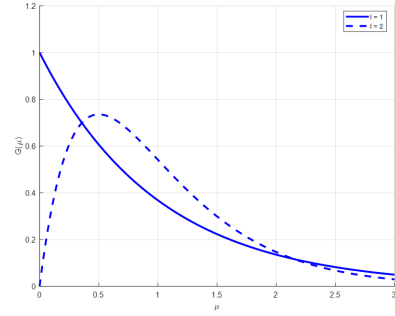


FIG. 1: Energy distribution $G(\mu)$ of hot electrons.

The normalized magnetic moment distribution of the hot electrons is modeled by a gamma distribution

$$G(\mu) = \frac{l^l}{\Gamma(l)} \mu^{l-1} \exp(-l\mu) \quad (42)$$

with $\Gamma(l)$ the gamma function. When $l = 1$, Eq. (42) represents a Maxwellian distribution of hot electrons. The normalized μ space is discretized with the normalization for the distribution $\sum_i G(\mu_i) \Delta\mu = 1$ where $\mu_i = i \Delta\mu$ with $i = 1, \dots, 5$ and an interval of $\Delta\mu = 0.25$. Figure 1 shows the distribution. For each μ value, we solve the Vlasov equation (22)

$$\frac{\partial F}{\partial t} + \frac{\partial}{\partial y} [(\hat{\omega}_d + v_y)F] + \frac{\partial}{\partial x} (v_x F) = S, \quad (43)$$

where $S(x, \mu) = S_0(x)G(\mu)$ represents the source term for hot electron production. The drive term

of the vorticity equation $\int d\mu \hat{\omega}_d F$ is calculated by $\sum_i \mu_i x^2 F(\mu_i) \Delta\mu$, once the F has been advanced in time. Fourier transforming the vorticity equation (34) in y gives

$$\left(-m^2 \frac{0.6}{x^4} + \frac{0.7}{x^2} \frac{\partial^2}{\partial x^2}\right) \dot{\phi}_m(x) = w_m(x), \quad (44)$$

where $w = \hat{\omega}_{ci0} \int d\mu \hat{\omega}_d \partial F / \partial y$. This equation can be inverted to obtain $\dot{\phi}_m(x)$ using a standard tridiagonal solver, and then inverse transformed in y to give $\dot{\phi}(x, y)$. To model non-resonant dissipation, we introduce a diffusion term in the potential update:

$$\phi(t + \Delta t) = \phi(t) + \Delta t \dot{\phi} + \nu \Delta t \nabla^2 \phi, \quad (45)$$

where ν is the diffusion coefficient.

We have developed an explicit, finite-difference code for solving Eqs. (43)–(45) on uniform, rectangular grids in the (x, y) plane. This code is a modification of our earlier code used for the study of interchange modes [27, 28]. The nonlinearities in Eq. (43) arise from the convective terms, which are written in conservative form. The gradients of the flux are evaluated to fourth order in the grid length and a velocity-dependent hyperviscosity is introduced for the convective term [29]. Equations (43) and (45) are advanced using a second-order trapezoidal leapfrog algorithm [29]. The simulation domain is radially global with the outer wall at $x_{min} = 0.3$ and the inner wall at $x_{max} = 1.5$. The radial boundaries are assumed to be conducting walls and therefore Dirichlet boundary conditions are imposed with $\Phi = 0$ and $F = 0.1$ at the outer and inner walls. Periodic boundary conditions are used for all quantities in the azimuthal direction for $y \in [0, 2\pi]$. The constant parameters used in the simulations are $\hat{\omega}_{ci0} = \omega_{ci0}/\omega_{d0} = 3.5$ [1], the diffusion coefficients $\nu = 0.002$, the time step $\Delta t = 0.01$. The hot electron source is radially localized with a Gaussian profile, which has a peak amplitude of $S_0 = 0.005$ and is centered at $x = 1$.

We now turn to nonlinear simulations of Eqs. (43)–(45) which show the self-consistent evolution of the HEI in dipole geometry. Both the observations and the previous simulations show that the fluctuations are dominated by the azimuthal mode number $m = 1$. Thus we focus on the $m = 1$ mode in the present simulation. Small perturbations are added to the potential. The initial hot electron profiles are uniform in y and radially localized at the source location. As shown in Fig. 3, the hot electron interchange modes are excited in the region where the radial gradient in F is sufficiently steep with $dF/dx > 0$. Here the magnitude of $F \propto \alpha$ essentially represents the fractional density of hot electrons. As predicted in the linear theory Eq. (41), the HEI modes are more unstable at low hot electron energy $\omega_d \propto \mu$ and high hot electron fraction α . The potential perturbations grow exponentially with azimuthal propagation in the electron magnetic drift (positive y) direction and develop small perpendicular mode number $m_\perp$ with broad radial mode

structures during the linear stage.

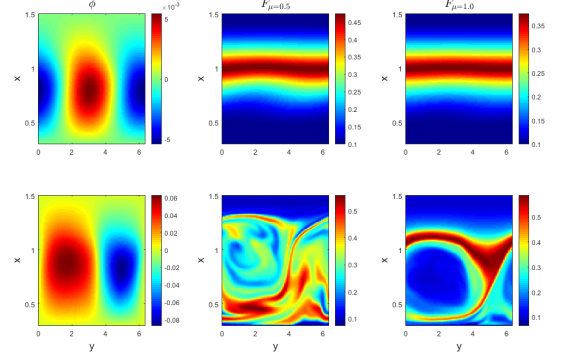


FIG. 2: Snapshot of phase-space structures of Φ and $F(\mu)$ with $\mu = 0.5$ and $\mu = 1$ in the linear (top) and nonlinear (bottom) phases.

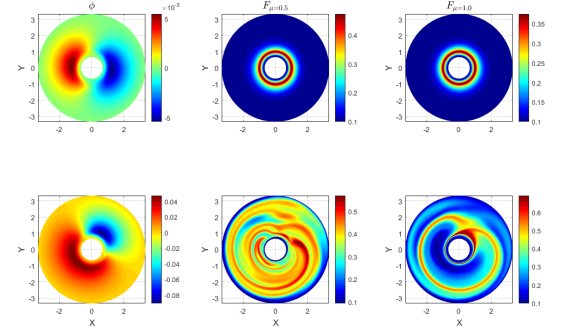


FIG. 3: Snapshot phase-space structures in the laboratory coordinates with normalized equatorial radius $L/L_0 = 1/x$ at the linear (top) and nonlinear (bottom) stages.

Figure 3 shows snapshots of the electrostatic potential and phase-space distribution during the linear growth stage ($t \approx 30$). The potential exhibits a regular $m = 1$ pattern, and the phase-space distribution shows periodic modulations but no coherent holes yet.

Figure 4 presents the nonlinear stage ($t \approx 150$). The potential develops localized structures, and clear phase-space “holes” (depressions in F_μ) are visible, which propagate inward and are responsible for the upward frequency chirping.

To further characterize the nonlinear transport, we compute the radial flux of hot electrons and average it over the poloidal direction y . Figure 5 shows the time evolution of this averaged flux as a function of the normalized radial coordinate x . The radial flux exhibits clear periodic oscillations, with the largest amplitude occurring during the linear stage ($t \lesssim 40$), indicating the most vigorous radial transport at that time. As the instability enters the nonlinear saturated phase, the flux amplitude

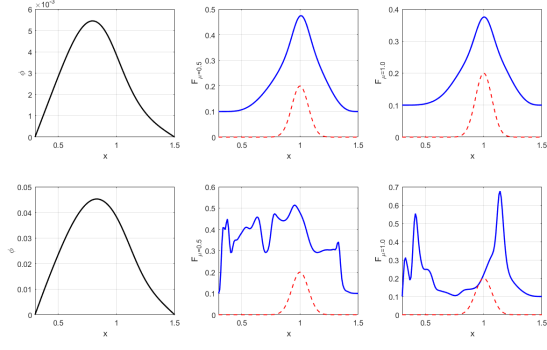


FIG. 4: Snapshot of radial profiles of Φ and $F(\mu)$ with $\mu = 0.5$ and $\mu = 1$ at a given azimuthal angle $y = \pi$ in the linear (top) and nonlinear (bottom) phases.

gradually decays. The flux is primarily localized near $x \approx 0.8$ and slowly shifts inward over time. Positive and negative values indicate outward and inward transport, respectively.

The averaged radial flux of hot electrons is defined by $\Gamma_x = \langle N_h v_x \rangle_y$ where the bracket $\langle f \rangle_y = \int f dy / 2\pi$ denotes the azimuthal average and $v_x = \partial\phi/\partial y$ represents the radial component of $\mathbf{E} \times \mathbf{B}$ motion. The hot electron number density is obtained by $N_h = \sum_i F(\mu_i) \Delta\mu$.

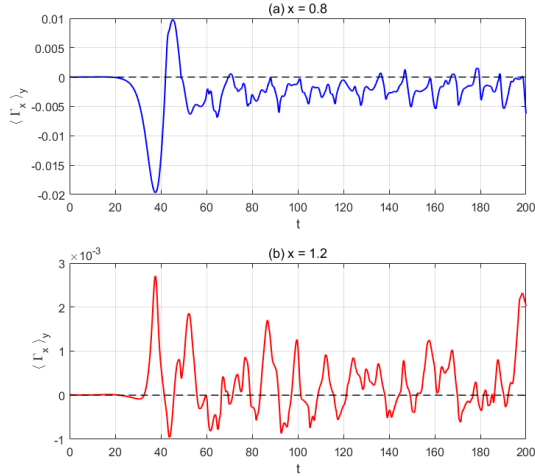


FIG. 5: Time evolution of the averaged radial flux $\langle \Gamma_x \rangle_y$ at different flux surface.

The instantaneous frequency is defined by the phase factor $\exp(-i \int \omega dt)$. Fourier transform of the potential $\phi(x, y, t)$ in y yields $\phi_m(x, t) = |\phi_m| \exp(i\theta_m)$ with the phase $\theta_m(x, t) \equiv \arctan(\Im\phi_m/\Re\phi_m)$. With the phase $\theta_m = -\int \omega dt$, the instantaneous frequency is obtained from the time derivative of the phase [30],

$$\omega_m(x, t) = -\frac{d}{dt}\theta_m(x, t). \quad (46)$$

Figure 6 presents the time evolution of the baseline case, including the phase of the $m = 1$ mode, instantaneous frequency, and the electrostatic potential at the plasma center.

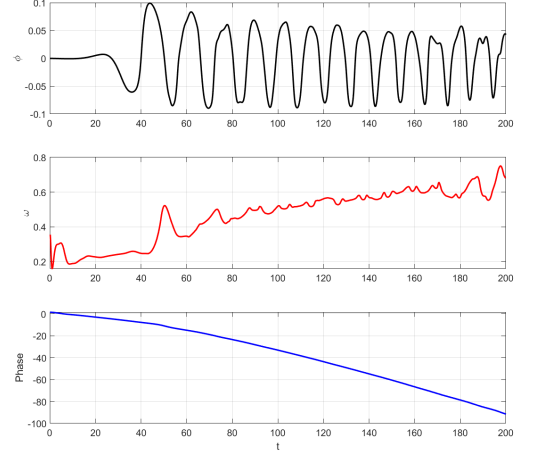


FIG. 6: Time evolution of the potential and frequency for $\nu = 0.002$.

The frequency chirping accompanying this evolution is shown in Fig. 7. The dominant frequency sweeps from ~ 0.5 to ~ 1.4 over the simulation period.

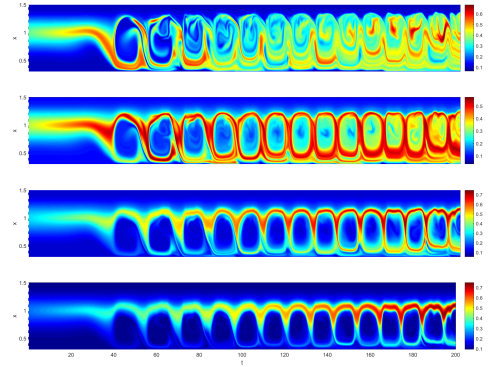


FIG. 7: Time evolution of F across the flux surfaces at a given y .

A. Role of the Particle Source

To isolate the role of the hot electron source, we performed a simulation with $\nu = 0.002$ but with the source term removed. The absence of a source leads to a natural decay of the instability as free energy is extracted. The maximum fluctuation amplitude decreases more rapidly than in the source-driven case, and the total frequency

chirp is reduced (final frequency ~ 1.0 vs. ~ 1.4). This confirms that the continuous particle source supplies free energy, allowing a wider frequency sweep.

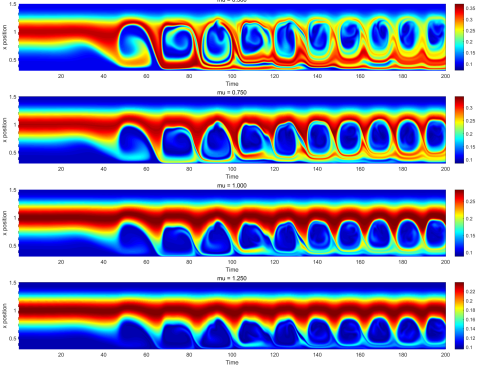


FIG. 8: Phase-space evolution in the absence of a particle source. Clear inward propagation of holes and amplitude decay are observed.

B. Energy-Dependence of Frequency Chirping: Single- μ Simulations

To determine which energy population drives the chirping, we performed simulations with only a single magnetic moment ($\mu = 0.5$ or $\mu = 1.0$). Figure 9 compares the frequency chirping for these two cases. Both exhibit chirping independently, but the higher-energy ($\mu = 1.0$) case produces a much stronger frequency sweep, comparable to the full multi- μ case. The chirp onset time is nearly identical. This indicates that while low-energy electrons contribute to initial turbulence, the high-energy population can independently generate and sustain a strong frequency chirp through resonant interaction.

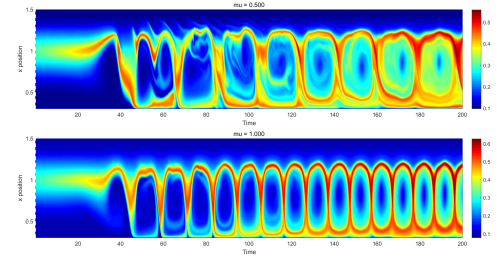


FIG. 9: Comparison of frequency chirping in single- μ simulations. Both $\mu = 0.5$ and $\mu = 1.0$ show chirping, with the higher energy case exhibiting a much stronger effect.

VI. CONCLUSION

The self-consistent nonlinear simulations have been carried out. The nonlinear model reproduces the frequency sweeping observed experimentally. The creation and dynamical evolution of phase-space structures are essential to the nonlinear saturation of the HEI.

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Appendix A: Geometric integrals for a point dipole

In the derivation of the flux-tube averaged Poisson equation, we have the following two geometric integrals:

$$I_1 = \int \frac{d\chi}{B^2} R^2 B^2, \quad I_2 = \int \frac{d\chi}{B^2} \frac{1}{R^2}. \quad (\text{A1})$$

For a point dipole in spherical coordinates (r, θ, ϕ) ,

$$B = \frac{M}{r^3} \sqrt{1 + 3 \cos^2 \theta}, \quad \psi = \frac{M \sin^2 \theta}{r}, \quad \chi = \frac{M \cos \theta}{r^2}. \quad (\text{A2})$$

A dipole field line is given by $r = L \sin^2 \theta$. Since

$$\frac{d\chi}{d\theta} = -\frac{M}{L^2} \frac{1 + 3 \cos^2 \theta}{\sin^5 \theta} \quad (\text{A3})$$

we obtain

$$\frac{d\chi}{B^2} = \frac{L^4}{M} \sin^7 \theta d\theta. \quad (\text{A4})$$

With $R = r \sin \theta = L \sin^3 \theta$, we have

$$R^2 B^2 = \frac{M^2}{L^2} (1 + 3 \cos^2 \theta) \sin^{-6} \theta. \quad (\text{A5})$$

Thus

$$I_1 = M \int_0^\pi \sin \theta (1 + 3 \cos^2 \theta) d\theta = 4M. \quad (\text{A6})$$

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